INDIFY

Product Requirements Document Sections 1–3

Institutional Cross-Border B2B Payment & Settlement Platform
Dubai / UAE ↔ India

Product	Indify
Category	Institutional Cross-Border Payments / Web3 Settlement Infrastructure
Primary Corridor	UAE ↔ India, with global expansion capability
Architecture	EVM-compatible Layer 2 or permissioned enterprise consortium network
Target Users	Corporate Treasurers, Import/Export Businesses, Regional Commercial Banks
Version	1.0
Date	October 2026

Document status: Product Definition / Architecture Baseline

Regulatory Architecture Note

Important: This document is a product and architecture specification, not a legal opinion or regulatory authorization. Indify should be designed around the exact licensing perimeter applicable to each activity and jurisdiction.

The UAE architecture must distinguish applicable VARA requirements from CBUAE payment-token requirements. CBUAE has a dedicated Payment Token Services Regulation covering relevant payment-token issuance, conversion, and custody/transfer services. The older Stored Value Facility framework should therefore not be treated as the sole regulatory basis for a tokenized-payment product.

For India, INR tokenized settlement must be structured around applicable RBI, FEMA, payment-system, banking and AML/KYC requirements and participating regulated entities. Regulatory authorization is a hard deployment gate.

SECTION 1 — EXECUTIVE SUMMARY, PROJECT BACKGROUND & HIGH-LEVEL ARCHITECTURE

1.1 Executive Summary

Indify is an institutional B2B cross-border payment and settlement platform designed to provide UAE-based businesses with an alternative settlement rail for international trade, initially focused on the UAE–India corridor.

The platform combines institutional digital wallets, regulated fiat/token conversion, EVM-based programmable settlement, AED/USD/INR-denominated settlement representations, enterprise smart-contract escrow, automated FX execution, liquidity-provider connectivity, corporate approval workflows, compliance/KYB/KYC, transaction monitoring, institutional custody/MPC, bank integrations, and an interoperability layer for conventional banking and future multi-CBDC environments.

Strategic principle: Indify does not seek to replace sovereign currencies, central banks, or regulated banks. It provides a programmable settlement layer connecting regulated financial institutions, corporates and tokenized-value networks.

1.2 Strategic Problem

Cross-border B2B payments can involve multiple correspondent banks, FX providers and settlement windows, creating multi-day settlement, prefunding requirements, FX spreads, intermediary fees, limited visibility, reconciliation overhead and trapped liquidity.

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Buyer → Originating Bank → Correspondent / Intermediary → FX Provider → Beneficiary Bank → Seller
```

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Indify target: Corporate → Indify → Regulated Liquidity / Settlement Network → Corporate
```

1.3 Product Vision

Build a regulated, programmable, interoperable institutional settlement network connecting UAE and India businesses to faster and more capital-efficient cross-border liquidity.

1.4 Product Goals

Goal	Product Requirement
Reduce USD dependency	Enable eligible alternative corridors such as AED → INR while retaining USD as a supported currency and reference currency.
Accelerate settlement	Target digital settlement finality ≤60 seconds under normal network/compliance conditions; distinguish blockchain finality from fiat finality.
Reduce intermediary costs	Minimize unnecessary correspondent fees, FX layers, prefunding and manual reconciliation.
Programmable settlement	Support escrow, milestone payments, invoice-linked settlement, multi-party settlement and automated reconciliation.

1.5 Target Audience Profiles

Persona	Needs / Primary KPI
Corporate Treasurer	Predictable settlement, liquidity visibility, FX execution, approvals and auditability. KPI: cost and time per settled transaction.
Dubai / DMCC Importer	AED→INR conversion, invoice-linked payment, escrow, supplier verification and documentary compliance.
Indian Exporter	Verified UAE counterparty, payment certainty, INR settlement, reconciliation and transparent FX.
Regional Commercial Bank	Institutional liquidity, regulated access, APIs, compliance data, settlement finality and reconciliation.
Compliance Officer	KYB/KYC, sanctions screening, AML monitoring, wallet screening, freeze controls and complete audit trail.

1.6 High-Level System Architecture

```
INSTITUTIONAL USERS
Corporate Treasurer | CFO | Finance Team | Bank | Admin
↓
INDIFY WEB APPLICATION
Dashboard | Wallet | FX | Payments | Escrow | Compliance
↓
APPLICATION / API ORCHESTRATION
Auth | RBAC | Workflow | Pricing | Risk | Compliance API
■ ■
GOOGLE AI STUDIO / GEMINI COMPLIANCE SERVICES
AI assistant KYB/KYC | AML | Sanctions
Document extraction Risk | Monitoring
Transaction explanation
↓
INDIFY WEB3 GATEWAY
RPC | Chain Abstraction | Tx Builder | Nonce | Gas | Simulation
↓
EVM L2 / PRIVATE CONSORTIUM
Wallet Controller | Payment Tokens | FX/AMM | Escrow
Settlement | Compliance Registry | Interoperability Adapter
↓
AED | INR | USD
↓
BANKS | FX PROVIDERS | PSPs | CBDCs | ISO 20022 APIs
```

1.7 AI Governance Boundary

Google AI Studio/Gemini is an intelligence and orchestration-assistance layer, not a financial authority. It may extract documents, summarize transactions, explain compliance outcomes and assist users. It must not independently sign transactions, release escrow, approve compliance exceptions, modify balances, change contract parameters or bypass corporate controls.

```
User → RBAC → Policy Engine → Compliance Engine → Corporate Approval → MPC/Multi-Sig → Web3 Gateway →
Smart Contract
```

1.8 Scope

In scope: institutional wallets; AED settlement; INR settlement; USD settlement; approved stablecoins/payment tokens; institutional FX; B2B settlement; escrow; programmable payments; invoice-linked settlement; MPC; multi-signature treasury; KYB/KYC; AML monitoring; wallet screening; automated freeze; EVM L2/private network; RPC gateway; blockchain indexer; bank APIs; ISO 20022 interoperability.

Out of scope: retail P2P trading; consumer remittances; anonymous wallets; permissionless token issuance; public token speculation; leveraged trading; derivatives; crypto lending; yield farming; retail DeFi; unlicensed money transmission; unrestricted self-hosted-wallet interaction; algorithmic stablecoin issuance.

SECTION 2 — FUNCTIONAL REQUIREMENTS & USER STORIES

Epic 1 — Multi-Currency Institutional Wallet

Objective: Provide corporations with secure institutional wallets capable of holding, receiving and initiating approved digital-value transactions across AED, USD and INR settlement representations, using MPC, multi-signature governance and role-based transaction policies.

US-1.1 — Create Institutional Wallet

As a Corporate Treasury Administrator, I want to create an institutional wallet, so that my organization can securely manage cross-border settlement assets.

Acceptance Criteria:

• Given the organization has completed KYB, when the administrator creates a wallet, then the platform shall generate an institutional wallet under the organization's legal entity.

• Given wallet creation is successful, when the wallet is provisioned, then the configured MPC/multi-signature policy shall be established.

• Given wallet creation is incomplete, when the user exits, then no partially initialized wallet shall become transaction-capable.

US-1.2 — Multi-Signature Approval

As a CFO, I want high-value transactions to require multiple approvals, so that no single employee can move corporate funds.

Acceptance Criteria:

• Given a 2-of-3 approval policy, when a transaction exceeds its threshold, then the transaction shall enter PENDING_APPROVAL.

• Given only one approver has approved, then the transaction shall not be broadcast.

• Given the required threshold is reached and all policy conditions pass, then the transaction may proceed to signing.

US-1.3 — Multi-Currency Balances

As a Corporate Treasurer, I want balances displayed in AED, USD, INR and counterparty currency, so that I can understand global liquidity.

Acceptance Criteria:

• Given the wallet contains multiple supported assets, when the dashboard loads, then each asset shall show token balance, native value, AED equivalent, USD equivalent, INR equivalent, valuation timestamp and FX source.

• Given an FX source is unavailable, then the last valid rate may be displayed only with an explicit stale-rate indicator.

US-1.4 — Wallet Transaction Policy

As a Treasury Administrator, I want configurable transaction policies, so that payment authority reflects internal governance.

Acceptance Criteria:

• Policies shall support amount thresholds, approved beneficiaries, currency restrictions, geographic restrictions, approval count, daily limits, velocity limits and cooling-off periods.

US-1.5 — MPC Recovery

As a Corporate Security Officer, I want institutional recovery procedures, so that loss of an operator credential does not permanently strand funds.

Acceptance Criteria:

• Given one MPC participant is unavailable, when recovery is initiated, then the configured recovery quorum shall be required.

• Given recovery quorum is not reached, then no signing authority shall be restored.

Epic 2 — Cross-Border Settlement Engine

The engine shall execute institutional cross-border transactions using approved tokenized settlement assets, liquidity pools, AMMs, RFQ, OTC liquidity or regulated liquidity providers.

US-2.1 — Initiate Cross-Border Payment

As a Corporate Treasurer, I want to initiate an AED-to-INR payment, so that my Indian supplier receives the required settlement value without unnecessary correspondent-bank intermediaries.

Acceptance Criteria:

• Given sender and beneficiary are approved and sufficient liquidity exists, when the payment is initiated, then the engine shall calculate FX rate, quoted INR amount, network fee, liquidity fee, platform fee and expected settlement time.

• When the user confirms, then the system shall create a unique settlement ID.

US-2.2 — Automated FX Execution

As a Treasurer, I want the platform to automatically execute approved FX routes, so that I receive predictable institutional pricing.

Acceptance Criteria:

• Given multiple routes are available, when a quote is requested, then the router shall evaluate price, liquidity depth, slippage, counterparty risk, compliance eligibility, latency and total cost.

• Then it shall select the best policy-compliant route.

US-2.3 — Atomic Settlement

As a Corporate Treasurer, I want the currency exchange and settlement to be atomic, so that I do not bear unilateral settlement risk.

Acceptance Criteria:

• Given both settlement legs are available, when the atomic settlement contract executes, then either both contractual transfers complete or neither completes.

• Given one leg fails, when the timeout expires, then the transaction shall revert or enter controlled recovery.

US-2.4 — Slippage Protection

As a Treasurer, I want maximum slippage controls, so that large B2B payments cannot execute at materially worse rates.

Acceptance Criteria:

• Given maximum slippage is 25 bps, when executable price deviates beyond 25 bps, then automatic execution shall stop and the system shall return PRICE_PROTECTION_TRIGGERED.

US-2.5 — Liquidity Insufficiency

As a Treasurer, I want the platform to detect insufficient liquidity, so that a large payment does not partially execute.

Acceptance Criteria:

• Given requested payment exceeds available executable liquidity, when a quote is generated, then the platform shall either split execution under approved policy or return INSUFFICIENT_LIQUIDITY.

• No partial settlement shall occur unless explicitly authorized.

Epic 3 — Project mBridge Compatibility Mock Gateway

This is an interoperability abstraction layer and test gateway, not a claim of official mBridge connectivity. It models how Indify could communicate with a multi-CBDC network or central-bank ledger. The gateway should support API-based integration and ISO 20022-oriented mapping.

US-3.1 — Interoperability Message

As an Integration Architect, I want blockchain settlement instructions converted into standardized interoperability messages, so that external payment systems can consume them.

Acceptance Criteria:

• Given an approved blockchain settlement exists, when the gateway receives the event, then it shall generate a message containing transaction ID, originating/beneficiary institution, currency, amount, settlement timestamp, purpose, compliance status, blockchain hash and unique message ID.

US-3.2 — ISO 20022 Mapping

As a Bank Integration Engineer, I want Indify transactions mapped to ISO 20022-compatible messages, so that existing financial infrastructure can consume settlement information.

Acceptance Criteria:

• Given a complete settlement instruction, when ISO 20022 serialization is requested, then a schema-valid message shall be generated.

• Given required fields are missing, serialization shall be rejected with missing-field diagnostics.

US-3.3 — CBDC Mock Ledger

As a Solution Architect, I want to simulate a central-bank ledger endpoint, so that Indify can test future CBDC connectivity without production central-bank access.

Acceptance Criteria:

• The mock environment shall support mint, burn, transfer, reserve, redemption, settlement confirmation, rejection, timeout and reversal.

US-3.4 — CBDC Settlement Failure

As a Settlement Operator, I want external ledger failures reflected accurately, so that users never see a rejected transaction as final.

Acceptance Criteria:

• Given an external CBDC ledger rejects a transaction, when rejection is received, then the status shall become EXTERNAL_SETTLEMENT_REJECTED and the UI shall not represent the transaction as final.

Epic 4 — Compliance, KYC/KYB & Transaction Monitoring

Indify shall use a jurisdiction-aware compliance engine rather than a single global rule set. VARA, CBUAE and India/RBI requirements shall be mapped to applicable activities and participating regulated entities.

US-4.1 — Corporate KYB

As a Corporate Administrator, I want to complete KYB digitally, so that my company can become eligible for institutional payments.

Acceptance Criteria:

• Given incorporation information is submitted, when KYB begins, then the system shall verify legal entity, registration, ownership, UBOs, directors, authorized signatories, address, business activity, sanctions exposure and applicable adverse-media signals.

• Given KYB fails, wallet activation shall remain blocked.

US-4.2 — Sanctions Screening

As a Compliance Officer, I want all counterparties screened against applicable sanctions lists, so that prohibited parties cannot transact.

Acceptance Criteria:

• Given a new counterparty is created, when screening executes, then applicable sanctions/watchlists shall be checked.

• Given a potential match exceeds the configured threshold, the counterparty shall enter COMPLIANCE_REVIEW.

US-4.3 — Transaction Monitoring

As a Compliance Officer, I want real-time transaction monitoring, so that suspicious patterns can be detected before settlement.

Acceptance Criteria:

• The monitoring engine shall evaluate transaction size, velocity, geography, counterparty risk, wallet risk, unusual conversion, structuring, rapid in/out movement, sanctions exposure, behavioral anomalies and invoice/payment patterns.

US-4.4 — Automated Freeze

As a Compliance Officer, I want automatic transaction freezing when critical risk conditions occur, so that potentially illicit funds cannot settle.

Acceptance Criteria:

• Given a transaction generates a critical compliance alert, when the risk threshold is reached, then the transaction shall enter FROZEN_COMPLIANCE_REVIEW.

• Blockchain execution shall be blocked where technically possible; off-chain settlement shall be blocked; users receive a generic compliance status; compliance receives the detailed alert.

US-4.5 — Audit Trail

As a Compliance Officer, I want every compliance decision recorded, so that decisions are reviewable and auditable.

Acceptance Criteria:

• Each decision shall record actor, timestamp, rule version, decision, risk score, supporting evidence and override reason/authority where applicable.

Epic 5 — Seamless USD / INR / AED Conversion

US-5.1 — Currency Conversion

As a Corporate Treasurer, I want to convert between AED, USD and INR, so that I can manage settlement currency without multiple external systems.

Acceptance Criteria:

• Given the user holds AED and requests AED→INR, then the system shall display executable rate, source, timestamp, gross amount, destination amount, spread, fees and expected settlement time.

US-5.2 — FX Quote Expiry

As a Corporate Treasurer, I want expired quotes rejected, so that I cannot unknowingly execute stale pricing.

Acceptance Criteria:

• Given a quote has a 15-second validity period, when it expires, then execution shall require a new quote.

US-5.3 — Large-Value FX

As a Corporate Treasurer, I want large-value FX routed through institutional liquidity, so that execution reflects enterprise pricing and liquidity.

Acceptance Criteria:

• Given the transaction exceeds the standard threshold, when conversion is requested, then the platform shall use an institutional RFQ, OTC, institutional AMM, bank liquidity or regulated payment-token conversion route.

SECTION 3 — NON-FUNCTIONAL REQUIREMENTS & SECURITY / COMPLIANCE FRAMEWORK

3.1 Regulatory Architecture

Activity	UAE Domain	India Domain
Virtual-asset activity	VARA where applicable	Applicable Indian framework
Payment token	CBUAE	RBI/FEMA/banking framework
Custody	VARA / CBUAE depending on activity	Applicable regulated entity
Fiat settlement	UAE licensed financial institution	Indian regulated bank/payment institution
INR conversion	UAE/India licensed counterparties	RBI/FEMA requirements
AML/CFT	UAE AML/CFT + applicable regulator	PMLA/RBI requirements
Corporate KYC	UAE requirements	RBI/regulated-bank requirements
Cross-border transaction	UAE payments/FX rules	FEMA/payment-system requirements

3.2 VARA Compliance Requirements

• Technology risk assessment and governance.

• Cybersecurity controls, cryptographic key management and wallet-management controls.

• Testing, audit and transaction-control processes.

• AML/CFT, customer risk classification, sanctions screening, monitoring and regulatory reporting.

• Market-conduct controls including transparent fees, disclosures, complaints and customer agreements.

3.3 CBUAE Compliance

Indify's operating model must determine whether Indify itself, a banking partner or another licensed entity performs each regulated function. The design must explicitly map issuance, conversion, custody/transfer, payment services and fiat settlement to the relevant authorization perimeter.

3.4 India / RBI Compliance

Indify shall not treat an INR token representation as automatically equivalent to permission to issue or operate an INR stablecoin. The India model shall be built around applicable RBI requirements, FEMA, payment-system authorization, PMLA, KYC, reporting, authorized-dealer/banking relationships and other applicable restrictions.

3.5 Security Architecture

• MPC and HSM-backed key components.

• Encrypted key shares and role-separated access.

• Quorum-based signing and hardware-backed authentication.

• Key rotation, emergency revocation and disaster recovery.

• Geographically separated key components and controlled operational access.

3.6 Transaction Signing Architecture

```
User
↓
Authentication
↓
RBAC
↓
Corporate Policy Engine
↓
```

```
Compliance Screening
↓
Risk Engine
↓
Transaction Simulation
↓
Corporate Approval / Multi-Sig
↓
MPC Authorization
↓
HSM / Secure Signer
↓
Web3 Gateway
↓
EVM Network
```

3.7 Smart Contract Security

Every production contract shall undergo threat modeling, static analysis, automated testing, fuzz testing, invariant testing, economic attack analysis, reentrancy/access-control/oracle analysis and upgradeability review.

External audit minimum: two independent specialist smart-contract security audits before production deployment for material contracts. High-risk contracts should additionally undergo formal verification where practical, economic-security review, adversarial testing and a bug bounty.

3.8 Smart Contract Governance

• Controlled upgradeability where legally and technically appropriate.

• Multi-signature upgrade authority.

• Timelock for non-emergency changes.

• Emergency pause and controlled unpause.

• Version registry and upgrade proposals.

• On-chain governance event logging.

• Emergency role revocation and oracle freeze.

3.9 Performance Requirements

Metric	MVP Target	Production Target
Blockchain throughput	≥500 TPS	≥2,000 TPS
API throughput	≥1,000 req/s	≥5,000 req/s
Payment initiation API	<500 ms	<250 ms
Quote generation	<2 sec	<1 sec
Compliance pre-screen	<5 sec	<3 sec
Transaction submission	<2 sec	<1 sec
Digital settlement finality	≤60 sec	≤15 sec target
Dashboard balance refresh	<5 sec	<2 sec
System availability	99.9%	99.99%

These are product engineering targets for an L2/enterprise network, not claims about Ethereum mainnet performance.

3.10 Latency Requirements

• Application latency: ≤500 ms target for ordinary API operations.

• Compliance latency: ≤3 seconds target for preliminary risk result.

• Blockchain latency: ≤15 seconds target for confirmation on production enterprise/L2 network.

• Fiat settlement latency: dependent on external banking/payment infrastructure.

• UI must not equate blockchain confirmation with fiat finality unless the relevant fiat leg has settled.

3.11 Availability & Resilience

• Multi-zone deployment and redundant RPC nodes.

• Redundant blockchain indexers and durable message queues.

• Database replication and automated failover.

• Business continuity and disaster recovery.

• RTO ≤30 minutes.

• RPO ≤5 minutes; critical financial state should approach zero through synchronous/quorum replication where justified.

3.12 Oracle & FX Security

```
Bank FX Feed
+
Institutional Liquidity Provider
+
Independent Market Reference
↓
Price Aggregator
↓
Deviation Detector
↓
Execution Price
```

Execution shall halt when oracle deviation exceeds policy, liquidity disappears, pricing becomes stale, abnormal volatility occurs or price-source quorum is lost.

3.13 Compliance Freeze Architecture

```
CREATED
↓
KYC/KYB_CHECK
↓
RISK_SCREEN
↓
COMPLIANCE_APPROVED
↓
CORPORATE_APPROVAL
↓
SIGNED
↓
BROADCAST
↓
CONFIRMED
↓
SETTLED

ANY STATE → RISK ALERT → FROZEN → COMPLIANCE REVIEW
■■ RELEASE
■■ REJECT
■■ ESCALATE
```

3.14 Automated Freeze Triggers

• Sanctions match or blocked jurisdiction.

• High-risk counterparty or wallet.

• Compromised wallet indicators.

• Abnormal transaction velocity or structuring.

• Transaction-policy violation.

• Oracle manipulation or smart-contract exploit.

• Compliance-system outage where fail-closed policy applies.

• External regulatory instruction.

3.15 Identity & Authentication

• Corporate SSO, OAuth/OIDC, hardware security keys and MFA.

• Role-based access control and device binding.

• UAE PASS integration path for corporate identity verification/authentication where technically and legally available.

• UAE PASS identity authentication must remain distinct from corporate KYB/UBO verification.

3.16 Localization

Requirement	Specification
English	Primary operational language.
Arabic	Required for UAE deployment; full RTL support.
Hindi	Required for India-oriented user experience.
Dates / numbers	Localized formatting.
Compliance messaging	Localized transaction statuses, notices and disclosures.
Reporting currency	AED default for UAE corporate management reporting; USD, INR and transaction/counterparty currency also supported.

3.17 Audit & Data Integrity

Every financially material event must contain an immutable or tamper-evident event ID, timestamp, user ID, organization ID, transaction ID, wallet ID, blockchain hash where applicable, approval record, compliance decision, FX quote and execution price, and complete fee breakdown.

3.18 Privacy Architecture

The blockchain should not contain unnecessary personally identifiable information. On-chain records should be limited to settlement identifiers, wallet addresses, asset/amount/timestamp and cryptographic references. KYC documents, UBO data, sanctions evidence, invoices and confidential banking information should remain off-chain under controlled access.

3.19 AI Governance

Gemini/Google AI Studio integration shall be governed by a dedicated AI Risk Control Layer. AI may extract documents, classify invoices, summarize transactions, explain compliance decisions and identify anomalies for human review. AI may not independently release funds, approve transactions, modify balances, override sanctions, change compliance rules, authorize wallet signing or modify smart-contract governance.

3.20 Core System KPIs

Category	KPI	Target
Commercial	Settlement time	≤60 sec digital target
Commercial	Cost reduction	≥30% vs comparable correspondent route
Commercial	FX spread	≤50 bps for standard institutional flow
Commercial	Failed settlement rate	<0.1%
Commercial	Reconciliation automation	>95%

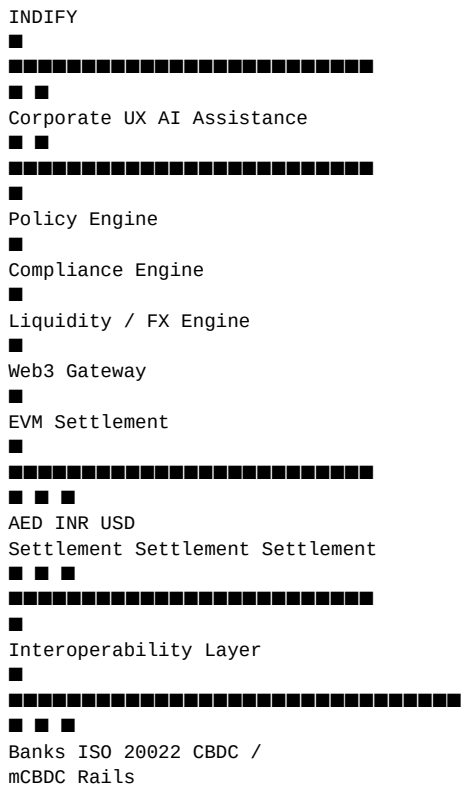
Category	KPI	Target
Operational	Availability	≥99.99%
Operational	Transaction monitoring coverage	100%
Operational	KYB automation	≥80%
Operational	Manual reconciliation	<5%
Operational	Critical security incidents	0

3.21 MVP Release Boundary

MVP includes: institutional onboarding, KYB/KYC, corporate wallets, MPC, multi-signature approvals, AED/USD/INR display, AED↔INR conversion, USD conversion, institutional settlement, escrow, compliance engine, transaction monitoring, automated freeze, audited smart contracts, Web3 gateway, banking API abstraction, mock multi-CBDC gateway, ISO 20022 mapping, English/Arabic/Hindi and UAE PASS integration path.

MVP excludes: retail users, consumer remittances, unrestricted crypto trading, anonymous wallets, public permissionless liquidity, algorithmic stablecoins, leverage, derivatives and yield products.

3.22 Architectural North Star



Reference Regulatory / Technical Sources

VARA — Virtual Asset Rulebooks: rulebooks.vara.ae

CBUAE — Payment Token Services Regulation: rulebook.centralbank.ae

BIS — Project mBridge: bis.org/project/mbridge

RBI — KYC Directions / amendments: rbi.org.in

These sources should be re-validated by Indify counsel and compliance leadership before each production release because regulatory requirements and technical specifications can change.